Version	Contributor	Date	Change Log
v1.0	kevin.liu	18/04/24	Publish

Overview

System events service is for apps on robots acquiring system's lastest status 、key events & publishing system events.

EventTypes

SysStatusType、BatteryType、KeyEventType-HeadEvent、KeyEventType-PowerEvent、KeyEventType-VolumnEvent etc.

Event Distribution Strategy

- All push
- PriorityOrder push

Event Subscribe & Unsubscribe

Subsribe event receiver in need & unsubscribe in time.

SysEventApiImpl.class API Description

```
    /**
     * subscribe event receiver
     *
     * @param event which extends BaseEvent see@EventType & ReceiverTpe
     * @param receiver which extends BaseReceiver see@EventType & ReceiverTpe
     */
    void subscribe(final BaseEvent event, final BaseReceiver receiver);

    /**
     * unsubscribe event receiver
     *
     * @param receiver which extends BaseReceiver see@EventType & ReceiverTpe
     */
    void unsubscribe(final BaseReceiver receiver);

    /**
     * unsubscribe by skill
     * @param skill
     */
    void unsubscribeBySkill(MasterSkill skill);

    /**
     * async get battery info
     * @param getBatteryInfoListener async callback
     */
    void getCurrentBatteryInfo(final GetBatteryInfoListener getBatteryInfoListener);

    /**
     * sync get battery info
     * @return battery info
     */
   SysMasterEvent.BatteryStatusData getCurrentBatteryInfoSync();
    /**
     * async get system status
     * @param getSysActivieStatusListener async callback
     */
    void getCurrentSysActiveStatus(final GetSysActivieStatusListener
getSysActivieStatusListener);

    /**
     * sync get system status
     * @return 系统状态
     */
    SysMasterEvent.ActiveStatusData getCurrentSysActiveStatusSync();

    /**
     * async publish
     * @param newStatus new system status
     * @param listener publish callback
     */
```

```
    void publishSysActiveStatus(SysMasterEvent.ActivieStatusType newStatus, @NonNull final
PublishSysActiveStatusListener listener);

    /**
     * sync publish
     * @param newStatus  new system status
     */
    void publishSysActiveStatusSync(SysMasterEvent.ActivieStatusType newStatus);
```

EventType & ReceiverTpe

events under event/ folder

- Key Type Events ：BatteryEvent、HeadEvent、PowerButtonEvent
- System Status Events & Battery Status Events 。

Tips：
All implmented event types have newInstance methods can make event object,which can take skill instance

receivers under receiver/ folder

- KeyEventReceiver Key Event Receivers extends BaseReceiver.
- BatteryEventReceiver/SysActiveStatusEventReceiver Status Receivers extends BaseReceiver.

SDK Use Samples

eg:

```
HeadEvent event = HeadEvent.newInstance(skill);
SingleClickReceiver receiver = new SingleClickReceiver(){
    @override
     public boolean onSingleClick(KeyEvent event){
           Log.d("SingleClickReceiver","onSingleClicked:"+event);
     }
};
SysEventApi.get().subscribe(event, receiver);
SysEventApi.get().unsubscribe(receiver);
```